FROSTBYTE: YOUTH WEEK HACKATHON 2026

"Break the Ice. Build the Future."

ELIMINATION ROUND — PROJECT REQUIREMENT SUBMISSION TEMPLATE

A. Team Information

Team Name:

FOUR+ONE

Team Roster (5 members):

#	Full Name	Role in the Team	Contact Number
1	BALINADO, Christian V.	Team Leader / Frontend Dev	09759610432
2	PADILLA, Rheden N.	Backend / Data Engine Dev	09942573523
3	VICTORIO, Emmanuel N.	UI/UX & Frontend Dev	09276786457
4	CARTECIANO, Adam Jeck G.	Taxonomy/Backend Dev	09395629350
5	BALOIS, Joshua Emmanuel T.	Research & Pitch Lead	09282854948

B. Project Brief

Project Title:

RoarCast - *Powering the Triple Helix of Santa Rosa*

Problem Statement (What real community problem are you solving?):

National unemployment has settled at around 4.7% (DOLE, 2026), but that number hides a real problem: youth underemployment and NEET (Not in Employment, Education, or Training) rates haven't moved much, because the timing is broken. Industries change what skills they need almost in real time. Curricula, at best, get updated once every few years. And PESO only steps in after the damage is already done — once a graduate shows up jobless.

Santa Rosa feels this acutely. The city's PEZA-registered zones — Laguna Technopark, Technopark Annex, Laguna International Industrial Park, Greenfield Automotive Park, and the Toyota Sta. Rosa Special Economic Zone — need people who know niche technical skills that local graduates simply were never taught in school.

This isn't a theory we're proposing a fix for. This disconnect was formally flagged on July 2, 2026 at the Laguna ASPIA (Association of PESO, Industry, and Academe) General Assembly in Santa Rosa City, where Mayor Arlene B. Arcillas and DOLE CALABARZON Regional Director Atty. Roy L. Tolentino called for a stronger Triple Helix (Government-Industry-Academe) partnership (Philippine Information Agency, 2026). Root-cause analysis traces the failure to one point: there is no real-time, autonomous intelligence-sharing infrastructure translating industry demand into academic supply — the Triple Helix is policy-supported but operationally disconnected. Existing tools like PhilJobNet and physical job fairs are reactive publication tools, not proactive preparation tools, and do not solve this timing problem.

Everyone in the loop pays for this. Students find out what they're missing only after they've already graduated and shown up to a PESO job fair — by then, it's lost time and lost income. Schools have no visibility into which specific skills their own graduates are lacking, so they can't adjust anything mid-stream. And the locators (companies/employers) end up either spending heavily to retrain people themselves or hiring from outside Laguna entirely — money and talent that should be staying in Santa Rosa, leaking out instead.

Proposed Solution (What is your project and how does it work?):

RoarCast, a web platform that actually connects Santa Rosa's PEZA ecosystem to the students trying to enter it, instead of leaving them to find out the hard way after graduation.

The name reflects what it does. "Roar" nods to Santa Rosa's own "Lion City" identity, but "Cast" is the functional part: the platform broadcasts real industry demand outward, giving students a read on where the job market is heading before they graduate into it, not after.

RoarCast runs on two simple roles. **Students** get the frontline experience: an automated Data Engine pulls job postings and skill requirements from Santa Rosa's PEZA-registered locators — across all the city's zones, not just one — and cross-checks them against official DOLE/TESDA regional forecasts, turning raw postings into structured Skill Tags. A student takes a 60-second micro-audit, sees exactly where their skills fall short against those tags, and joins a peer "Co-Op Upskilling Squad" to close the gap together, since solo online learners tend to disengage fast. Finishing a module earns a verifiable digital credential, built on the same standards behind W3C Verifiable Credentials and 1EdTech Open Badges, with a QR code anyone (a recruiter, a PESO officer) can scan to confirm it's real.

Admins (academe and LGU/PESO users sharing one role, one login, one dashboard) see the same aggregated, anonymized data through different filters: schools track where their curricula are falling behind, and PESO tracks where Special Education Fund money would close the widest gaps.

Target Users / Beneficiaries:

- **Primary:** Santa Rosa youth aged 16–24, especially graduating Senior High School and college students entering the local labor market.
- **Secondary:** Academe and Government (PESO/CYDO), represented through RoarCast's Admin role, using aggregated data for curriculum planning and fund allocation.

UN Sustainable Development Goal (SDG) Alignment:

- **SDG 8: Decent Work and Economic Growth (Target 8.6)** — *Primary Alignment*. Reduces youth unemployment and skills mismatch by aligning students with real-time industry skill demands, verified micro-credentials, and employment pathways before graduation.
- **SDG 4: Quality Education (Target 4.4)** — *Secondary Alignment*. Strengthens technical and vocational education by providing academe with real-time workforce intelligence to inform curriculum enhancement, micro-credential development, and industry-responsive skills training.

Center of Youth Participation Alignment:

- **Economic Empowerment (primary)** — integrates youth into the PEZA-driven economy through transparent skill requirements, peer Skill Squads, and verifiable credentials.
- **Governance (secondary)** — anonymized audit data informs PESO and City Government allocation of the Special Education Fund (SEF).

Team Roles and Contributions (who did what):

- **Balinado, Christian V. — Team Leader / Frontend Dev.**

  Leads overall project direction and coordinates the team; builds the frontend in React/Next.js and Tailwind CSS, ensuring an accessibility-compliant, frictionless student-facing interface.

- **Padilla, Rheden N. — Backend / Data Engine Dev.**

  Develops the Autonomous Data Engine on Firebase/Supabase, handling job-post ingestion, parsing, and the backend logic that turns raw postings into structured Skill Tags.

- **Victorio, Emmanuel N. — UI/UX & Frontend Dev.**

  Designs and builds the micro-audit flow, Co-Op Squad matching interface, and live dashboard views for Academe and PESO, focusing on usability for both students and admins.

- **Carteciano, Adam Jeck G. — Taxonomy/Backend Dev.**

  Builds and maintains the Dynamic Skill Tag taxonomy, mapping industry postings and DOLE/TESDA data to standardized skill categories used across the audit and matching engine.

- **Balois, Joshua Emmanuel T. — Research & Pitch Lead.**

  Leads research and sourcing (local context, SDG/CYP alignment, citations), and shapes the pitch narrative and documentation for judges.

F. Team Leader Certification

We certify that this project was conceptualized and developed within the official hackathon period and complies with the FROSTBYTE 2026 Hackathon Rulebook.

Team Leader Name and Signature:

SGD.
CHRISTIAN V. BALINADO

Date Submitted:
07 / 23 / 2026
